1. Problem statement

Develop a Python program to check scholarship eligibility using aggregate marks, subject-wise minimum marks, attendance, and family income. Rank eligible students according to their marks and resolve ties using attendance.

2. Objectives

* Check eligibility conditions.
* Identify students who qualify for scholarships.
* Rank eligible students.
* Resolve ties using predefined rules.

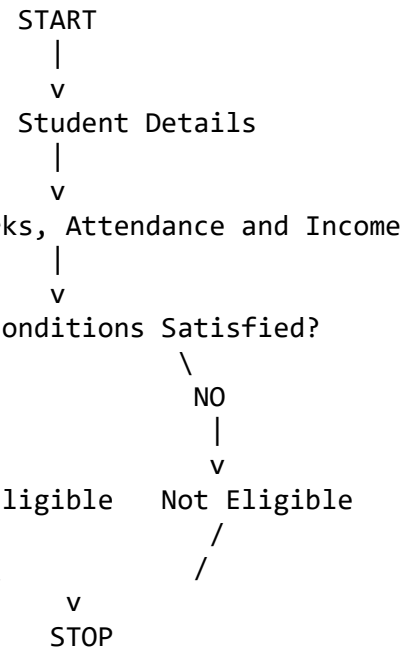
3. Assumptions

* Aggregate marks must be at least 80%.
* Each subject mark must be at least 50%.
* Attendance must be at least 75%.
* Annual family income must not exceed ₹2,50,000.
* Higher aggregate marks receive a better rank. If marks are equal, higher attendance wins.

4. Algorithm

* Start.
* Input student name, aggregate marks, subject marks, attendance, and family income.
* Check all eligibility conditions.
* If all conditions are satisfied, declare the student eligible.
* Otherwise, declare the student ineligible.
* Display the result.
* Stop.

5. Flowchart



6. Pseudocode

```
START
INPUT name, aggregate, subject marks, attendance, income

IF aggregate >= 80 AND
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   every subject mark >= 50 AND
   attendance >= 75 AND income <= 250000
    PRINT "Eligible"
ELSE
    PRINT "Not Eligible"
END IF
```

7. Python source code

```
name = input("Enter student name: ")
aggregate = float(input("Enter aggregate percentage: "))
subjects = list(map(float, input("Enter 3 subject marks: ").split()))
attendance = float(input("Enter attendance percentage: "))
income = float(input("Enter annual family income: "))

if (aggregate >= 80 and min(subjects) >= 50
        and attendance >= 75 and income <= 250000):
    print("Student:", name)
    print("Scholarship Eligible")
else:
    print("Student:", name)
    print("Scholarship Not Eligible")


STOP
```

8. TEST CASE

Test	Aggregate		Minimum subject mark	Attendance	Income	Expected result
1	90	75	90	₹2,00,000	Eligible	
2	70	65	90	₹2,00,000	Not eligible	
3	85	45	90	₹2,00,000	Not eligible	
4	85	65	80	₹3,00,000	Not eligible	